Department of Computer Science,

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Program: BSSE

LAB TASK-

Information Security

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Question no1:
Suggest three programming improvements in the above Diffie-Hellman code. Rewrite the updated program by applying these changes, and ensure that the modified code produces correct output. Also explain briefly how each change improves the program.

Solution:

```python
def power(a, b, p):
    if b == 1:
        return a
    else:
        return pow(a, b) % p

def main():
    P = int(input("Enter prime number P: "))
    G = int(input("Enter primitive root G: "))

    a = int(input("Enter Alice private key: "))
    b = int(input("Enter Bob private key: "))

    print("\n--- Public Values ---")
    print("P =", P)
    print("G =", G)

    x = pow(G, a, P)
    y = pow(G, b, P)

    print("\n--- Generated Keys ---")
    print("Alice sends:", x)
    print("Bob sends:", y)

    ka = pow(y, a, P)
    kb = pow(x, b, P)

    print("\n--- Secret Keys ---")
    print("Alice secret key:", ka)
    print("Bob secret key:", kb)
```

```python
if __name__ == "__main__":
    main()
```

Output:

```
Enter prime number P: 7
Enter primitive root G: 8
Enter Alice private key: 13
Enter Bob private key: 35

--- Public Values ---
P = 7
G = 8

--- Generated Keys ---
Alice sends: 1
Bob sends: 1

--- Secret Keys ---
Alice secret key: 1
Bob secret key: 1
PS F:\VS code folder ( IS_ Projects)>
```

Description:

This program is called Diffie Hellman key exchange. It is used to create a shared secret key between two people named Alice and Bob without sending the key directly.

First, both Alice and Bob agree on two public values. These are a prime number P and a base number G. Everyone can see these values.

Then Alice chooses her own secret number called a. Bob also chooses his own secret number called b. These values are private and are not shared with anyone.

Next, Alice uses the formula G raised to the power a modulo P to create a new value x. Bob uses the formula G raised to the power b modulo P to create a new value y. They exchange these values with each other.

After that, Alice takes the value y received from Bob and applies her secret number a again using the same formula to create a secret key. Bob takes the value x received from Alice and applies his secret number b to create his secret key.

At the end, both Alice and Bob get the same secret key even though they never shared it directly.

The main idea is that public values are known to everyone, private values are kept secret, and mathematical operations allow both users to reach the same final key securely.